

	Solution	Marks	Remarks
10. (a) (i)	$226 - 206 = 20$ (multiple of 4 for α) ∴ 202 Pb is the end product	1M 1A	
(ii)	% undecayed Ra-226 left $\frac{N}{N_0} = e^{-\frac{6.02}{1600} \times 90}$ $\approx 97.857\%$ Accept 97.8 % to 98.0 %	1M 1A	
(iii)	% undecayed Ra-226 left $\frac{50}{1600} = \left(\frac{1}{2}\right)^{\frac{t}{1600}}$ $\approx 97.9\%$	1A	
(b) (i)	∴ random process	2	
(ii)	Some of the daughter products of Ra-226 are also radioactive and may emit β particles.	1A	
(iii)	Reason: weaker ionizing power of β and γ radiations - raise the source to a distance greater than the range of α (a few cm) will cease to have sparks. - insert a paper between the source and the gauze, sparks will cease to produce.	1A 1A	

	Solution	Marks	Remarks
7. (a) (i)	Increase the separation between the double slit and the screen, D .	1A	
(ii)	The separation of the bright dots on the screen becomes larger, thus the percentage error in its measurement is smaller.	1A	
(iii)	The angular position of the 2 nd order bright dot $\theta = \tan^{-1} \left(\frac{1.56 \times 10^{-2}}{1.40} \right) = 29.124053^\circ$ Grating spacing $d = 2.5 \times 10^{-6} \text{ m}$ Applying $d \sin \theta = n\lambda$, Wavelength $\lambda = \frac{2.5 \times 10^{-6} \times \sin 29.12^\circ}{2}$ $= 6.08378 \times 10^{-7} \text{ m}$ $\approx 608.4 \text{ nm}$ (∴ 608 nm)	1M 1M 1A	Accept $\lambda = (6.06 \text{ to } 6.10) \times 10^{-7} \text{ m}$
(b) (i)	The equation can only be applied for - $\lambda \ll a$ (i.e. wavelength $\ll$ separation of the two sources), OR λ is much smaller than a - $a \ll D$ (i.e. separation of the two sources $\ll$ distance to the screen and detector), OR a is much smaller than D	1A 1A	Note: the order of magnitude of the wavelength of sound is about 10^{-2} m
(ii)	For the 1 st order maximum, wavelength $\lambda =$ path difference $PB - PA$ $= \sqrt{(1+0.5)^2 + 2^2} - \sqrt{(1-0.5)^2 + 2^2}$ $= 2.5 - 2.06155281 = 0.43844719 \text{ m} \approx 0.438 \text{ m}$ speed of sound: $v = f\lambda = 750 \times 0.4384$ $\approx 328.835 \text{ m s}^{-1}$	1M 1M 1A	Accept $v = 328 \text{ m s}^{-1}$ to 330 m s^{-1}

	Solution	Marks	Remarks
5. (a) (i)	$m(5.0 \text{ cm}) = 50 \text{ g}$ (10.0 cm) $m = 100 \text{ g}$ or 0.1 kg	1M 1A	
(ii)	Counter-weight position: $10.0 \text{ cm} \pm 0.1 \text{ cm}$ Percentage error $= 100\% \times \frac{0.1}{10.0} = 1\%$ ∴ $m = 101 \text{ g}$ to 99 g i.e. maximum error $\approx \pm 1 \text{ g}$	1M 1A	
(b)	Spring balance reading $= mg = (0.1 \text{ kg})(9.81 \text{ N kg}^{-1})$ $= 0.981 \text{ N}$ (1.0 N for $g = 10 \text{ N kg}^{-1}$) $\approx 1.0 \text{ N}$	1M/1A	
(c) (i)	counter-weight position on beam balance the same spring balance reading reading increases	1A+1A	
(ii)	The beam balance would fail to work / to measure the mass of the load, as the apparent weight is zero (or weightless), the counter-weight can take any position.	1A	

	Solution	Marks	Remarks
6. (a)	Place the (cylindrical) lens on the piece of paper and trace its outline, use ray tracing method e.g. - direct and trace a light ray along / parallel to the principal axis - direct and trace a light ray towards the lens Direct another light ray parallel to the first one / principal axis (by shifting the ray box) and trace the path(s) of the (emerging) ray(s) on the paper. Extend (the path of) the emerging rays (backward) and locate the point of intersection (on the focal plane containing F). Measure the distance from the point of intersection (or F) to the centre of the lens, which gives the focal length of the lens. Source of error: Scale uncertainty/accuracy/precision of the plastic ruler OR Unable to mark the path correctly because of the thickness of the beam of light from the ray box OR The ray(s) is/are not parallel (to the principal axis) OR Any reasonable answer	1A 1A 1A 1A 1A	Or Using lens formula: - use tracing method, direct and trace light ray towards the principal axis towards the lens - extend (the path of) the emerging ray backward and locate the point of intersection (with the principal axis) - measure the corresponding object distance u and image distance v (on the principal axis) - sub. u and v into lens formula to find f Accept: Using diagram(s) to supplement / simplify the description.
(b)		5	

	Solution	Marks	Remarks
8. (a) (i)	To terminals X	1A	
(ii)	$P = IV$ $800 \text{ W} = I(220 \text{ V})$ $I = 3.636364 \text{ A}$ $\approx 3.64 \text{ A}$	1M 1A	
(iii)	$800 = \frac{V^2}{R} = \frac{50^2}{4R}$ $R = 75.625 \Omega$ $P_{\text{loop warm}} = \frac{V^2}{4R}$ $= 800 \left(\frac{1}{5} \right) = 160 \text{ W}$ Alternative (I) $800 = \frac{V^2}{R} = \frac{50^2}{4R}$ $R = 75.625 \Omega$ $P_{\text{loop warm}} = \frac{V^2}{4R}$ $= \frac{220^2}{4(75.625)}$ $= 160 \text{ W}$ Alternative (II) Power $\propto \frac{1}{\text{resistance}}$ $\frac{P_{\text{loop warm}}}{P_{\text{standing}}} = \frac{R_{\text{eq}}}{4R}$ $P_{\text{loop warm}} = P_{\text{standing}} \left(\frac{1}{4} + \frac{1}{4R} \right)^{-1}$ $= 160 \text{ W}$	1M 1M 1A 1A	
(b) (i)	Electrical energy (consumed)	3	
(ii)	(1) only the fuse blows (2) only the RCCB cuts off the supply	1A 1A	

	Solution	Marks	Remarks
3. (a)	If the maximum load is exceeded, the braking distance will increase. If the action provided remains the same, vehicles would not be able to stop in time in case of emergency (thus dangerous). OR A larger friction is required in braking the vehicle within the same distance, accident may occur if the brakes cannot provide such frictional forces.	1A 1A 1A	
(b) (i)	If the brakes are applied continuously, thermal energy generated will heat up the brake pads / brakes to too high a temperature that the brakes may fail.	1A	
(ii)	Let D be the distance travelled along the ramp. Kinetic energy of vehicle becomes its gravitational potential energy: $\frac{1}{2} m(25)^2 = m(9.81)(D \sin 30^\circ)$ $D = 63.710499 \text{ m}$ $\approx 63.7 \text{ m}$ (62.5 m for $g = 10 \text{ m s}^{-2}$)	1A	
(c)	$QF: v^2 = u^2 + 2ar$ $0^2 = 25^2 + 2(-9.81 \sin 30^\circ)D$ $D = 63.7 \text{ m}$	1M 1A	
4. (a) (i)	K.E. + P.E. $= \frac{1}{2} (0.3)(4)^2 + (0.3)(9.81)(0.2)$ $= 2.4 + 0.5586 = 2.9586 \text{ J}$ $\approx 2.99 \text{ J}$ (3.0 J for $g = 10 \text{ m s}^{-2}$)	1M+ 1A	
(ii)	As the spring gun is fixed, there is external force acting on the system / the gun, total momentum (of the spring gun and cannon ball) is not conserved.	1M 1A	
(b)	Vertical: $s = ut + \frac{1}{2} at^2$ $0 = (4 \sin 50^\circ) t_1 - \frac{1}{2} (9.81) t_1^2$ $t_1 = 0.624705 \text{ s}$ (0.612836 s for $g = 10 \text{ m s}^{-2}$) $\approx 0.625 \text{ s}$ (0.613 s for $g = 10 \text{ m s}^{-2}$) Horizontal: $R = 4 \cos 50^\circ \times t_1 = 4 \cos 50^\circ \times 0.625$ $= 1.66210 \text{ m}$ $\approx 1.61 \text{ m}$ (1.57 m for $g = 10 \text{ m s}^{-2}$)	1M 1A 1A	
(c)	t_c increases since the initial vertical velocity / component is greater.	1A	

Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its contents with care.

General Marking Instructions

1. It is very important that all markers should adhere as closely as possible to the marking scheme. In many cases, however, candidates may have obtained a correct answer by an alternative method not specified in the marking scheme. In general, a correct answer merits the *answer mark* allocated to that part, unless a particular method has been specified in the question.

In the marking scheme, alternative answers and marking guidelines are in rectangles.

2. In the marking scheme, answer marks or 'A' marks are awarded for a correct numerical answer with a unit. If the answer should be in km, then cm and m are considered to be wrong units.

3. In a question consisting of several parts each depending on the previous parts, method marks or 'M' marks are awarded to steps/methods or substitutions correctly deduced from previous answers.

4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

$$\text{Or } \frac{t_f}{2} = \frac{u \sin \theta}{g} = \frac{4 \sin 50^\circ}{9.81}$$

$$\text{Accept } t_f = 0.61 \text{ s to } 0.63 \text{ s}$$

$$\text{Or } R = u \cos \theta \times \frac{2(u \sin \theta)}{g}$$

$$\text{Accept } R = 1.56 \text{ m to } 1.62 \text{ m}$$

Or work done against vehicle's weight component along the ramp by its kinetic energy:

$$\frac{1}{2}mv^2 = mg \sin \theta \times D$$

mass of vehicle = m

$$\text{Note: } D \sin 30^\circ = 31.8552 \text{ m}$$

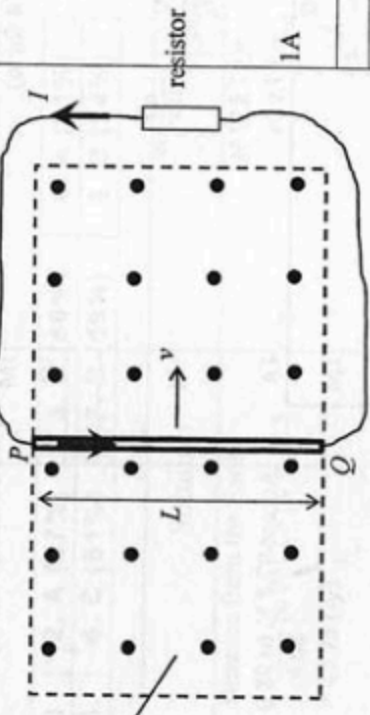
$$(\text{or } 31.25 \text{ m for } g = 10 \text{ m s}^{-2})$$

$$\text{Accept } D = 62.0 \text{ m to } 64.0 \text{ m}$$

$$= 2.4 + 0.6 = 3.0 \text{ J for } g = 10 \text{ m s}^{-2}$$

(%)	3. C (37%)	4. D (38%)
(%)	7. C (60%)	8. B (62%)

Solution	Marks	Remarks
ps: by heating a (tungsten) filament to a / red hot via joule heating (of a current), y becomes thermal energy / heat or only a converted to light output.	1A	
t sensitive to green light, ce (having the same light output power) ighter compared to a white one (comprises s).	1A	
each lamp:	1A	
$\cos \theta = \frac{10}{\sqrt{10^2 + 2.5^2}}$	IM	
$\theta \approx 14.0^\circ$	IM	
$\text{Or } \frac{E \times \cos \theta \times 2}{4\pi r^2}$	1A	
$\frac{000}{50} = 73.333333 \text{ lm W}^{-1}$	3	
$\frac{000}{35} = 74.074074 \text{ lm W}^{-1}$	1A	
needed.	1A	
illuminance smaller individual lamp failure smaller	1A	Any ONE
nt change / replacement of lamps ive as installation cost increases involved ation time	1A	Any ONE

Solution	Marks	Remarks
9. (a) (i) 	1A	
(ii) By Lenz's law, a magnetic force F_B acts on the rod such that it opposes the rod's motion, an external force F is needed to balance F_B so as to maintain uniform motion (or constant v) $\therefore F = F_B = ILB$ (in magnitude) <u>Or</u> $F \cdot v = I \xi$, $\therefore F = \frac{I \xi}{v}$	1A 1A 1A 1A	Accept : Work done by an external force is needed to transfer mechanical energy into electrical energy.
(iii) mechanical power input = Fv $= (ILB)v$ power input = power output (electrical) $ILBv = I \xi$ $\xi = BLv$	3 1M 1M	
(b) (i) Horizontal (component) is perpendicular to the mast. <u>Or</u> Vertical (component) is parallel to the mast.	2 1A 1A	
(ii) $\xi = (B \cos 30^\circ) L v$ $= (50 \times 10^{-6} \cos 30^\circ) (20) (6)$ $= 5.196152 \times 10^{-3} \text{ V}$ $\approx 5.20 \text{ mV}$ more electrons at end X	1 1M 1A 1A	by (a)(iii) using the component of B
(iii) No current. Both the cable and the mast cut the field lines in the same way, the e.m.f.'s produced are equal and thus oppose each other. <u>Or</u> No change in magnetic flux through the loop of the circuit.	3 1A 1A 2	

1. A (52%)	2. B (58%)
5. A (79%)	6. D (52%)

3. (a) (i) Incandescent lam

high temperature
most of the energy
small portion is c

(ii) As the eye is most
a green light sou
would appear brig
of different colour

(b) (i) Luminous flux of
 $E = 10000$ lumens

$$\tan \theta = \frac{2.5}{10} \text{ (or c)}$$

$$\theta = 14.036243$$

$$I = \frac{E \times \cos^3 \theta \times 2}{4\pi d^2}$$

$$= 14.532045 \text{ lux}$$

$$\approx 14.5 \text{ lux or lx}$$

$$\text{Efficacy (A)} = \frac{110}{1.5}$$

$$\text{Efficacy (B)} = \frac{100}{13}$$

Lamp B is recomm

(iii) Advantages:

- > Variation of
- > Effect of ind
- > Less glare

Disadvantages:

- > More frequen
- > More expens
- > More wiring
- > More installa

Section B : Atomic World

1. C (46%)	2. D (51%)	3. D (48%)	4. A (15%)
5. C (57%)	6. B (62%)	7. C (62%)	8. B (26%)

Solution	Marks	Remarks
2. (a) Increase / adjust the voltage just until there is no current passing through the circuit (no photoelectrons to complete the circuit), record the voltage V_s which gives the maximum kinetic energy of the photoelectrons $= eV_s$	1A 1A 2	
(b) (i) Ultra-violet (UV) radiation (light) (~278 nm)	1A	$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{10.8 \times 10^{14}} \approx 277.8 \text{ nm}$
(ii) slope of the graph = $\frac{3.3 - 0}{(13.4 - 5.4) \times 10^{14}}$ $= 4.125 \times 10^{-15} \text{ eV s } (= 6.6 \times 10^{-34} \text{ J s})$ The slope gives the Planck constant h .	1M 1A 1A 3	
(iii) Threshold frequency $f_0 = 5.4 \times 10^{14} \text{ Hz}$ Work function of sodium $= hf_0$ $= (6.6 \times 10^{-34}) \times (5.4 \times 10^{14})$ $= 3.564 \times 10^{-19} \text{ J}$ $= 2.2275 \text{ (eV)}$ $\text{Or } hf_0 = (4.125 \times 10^{-15}) \times (5.4 \times 10^{14})$ $= 2.2275 \text{ (eV)} \approx 2.23 \text{ (eV)}$	1M 1A	Accept 2.20 eV to 2.40 eV
(c) Unchanged, i.e. the same graph. The (maximum) kinetic energy depends on the energy of each photon, which is proportional to the frequency of radiation. Or Maximum kinetic energy of the photoelectrons / energy of photon 1A will not be affected.	1A 1A 2	